Question ID: ffdfe1e1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

There are 240 players in a tennis competition that includes 4 rounds of matches. Each player in the competition will play a match against another player in round 1. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p, eliminated at the end of round r, where $r \leq 4$?

Answer

A. $p = 15\left(\frac{1}{2}\right)^r$

B. $p = 15(2)^r$

C. $p = 240\left(\frac{1}{2}\right)^r$

D. $p = 240(2)^r$

Correct Answer: C

Rationale

Choice C is correct. It's given that one of the two players in a match is eliminated during each round. Therefore, the number of players decreases by half at the end of each round, and so the number of players eliminated at the end of a round also decreases by half. A decreasing exponential equation can be written in the form $p = a(1-b)^r$, where b is the value of the decrease each round, r, and a is the value of p, the number of players, when $r = 0$. It's given that there are 240 players in the tennis competition. Therefore, $a = 240$. Since the number of players eliminated at the end of a round decreases by half, it follows that $b = \frac{1}{2}$. Substituting 240 for a and $\frac{1}{2}$ for b in the equation $p = a(1-b)^r$ yields $p = 240\left(1-\frac{1}{2}\right)^r$, or $p = 240\left(\frac{1}{2}\right)^r$.

Choice A is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are 15, not 240, players.

Choice B is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are 15, not 240, players, and for which the number of players doubles, rather than decreases by half, after each round.

Choice D is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which the number of players doubles, rather than decreases by half, after each round.